IDENTIFYING KEY SOCIOECONOMIC DETERMINANTS OF INCOME INEQUALITY

A Machine Learning Approach Using World Bank Data

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Why it matters

- Rising inequality → reduced social mobility
- Political polarization
- Slower economic growth
- GINI coefficient varies significantly across countries and time
- Crucial for evidence-based policymaking

Persistent debates

- Which policy levers are most effective?
- Are drivers universal or context-dependent?
- Complex interactions between factors

Traditional econometric challenges

- **Non-linear relationships:** Complex interactions without explicit specification
- **High-dimensional data:** Multicollinearity & specification issues
- **Missing values:** Especially for developing countries
- Limited guidance on functional forms

ML advantages

Capture non-linearities | Handle high dimensions | Interpretable importance | Robust to noise

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RQ1: Feature Importance

Which socioeconomic indicators are the **strongest predictors** of GINI coefficients?

RQ2: Cross-Model Consistency

Are predictor rankings **consistent across different ML algorithms**?

RQ3: Predictive Performance

How well can we **predict inequality** across different contexts?

Contribution

Data-driven approach using **5 ML algorithms** to explore inequality drivers while implementing advanced optimization techniques

Data: World Bank Database (2000-2023)

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Target Variable

GINI coefficient | $\sim$**1,800** country-year observations

62 Predictor Variables

1. **Economic** (GDP, trade, inflation)
2. **Demographics** (population, urbanization)
3. **Human development** (health, education)
4. **Labor market** (employment rates)
5. **Infrastructure** (electricity, internet)
6. **Governance** (gender parity)

53 World Bank indicators + 9 engineered features

Preprocessing Pipeline

- KNN imputation (k=5)
- Exclude features >50% missing
- 80-20 train-test split
- 5-fold cross-validation

Training (80%)	Test

Five Tree-Based Algorithms

Data & Methodology

Model	**Type**	**Key Feature**
Decision Tree	Baseline	Single tree, recursive partitioning
Random Forest	Ensemble (Bagging)	Bootstrap + random features Reduces variance
Gradient Boosting	Ensemble (Boosting)	Sequential fitting of residuals Reduces bias
XGBoost	Advanced Boosting	Regularized objective 2nd-order optimization
LightGBM	Advanced Boosting	Histogram-based + GOSS Computational efficiency

Hyperparameters: 200 estimators, learning rate 0.05, max depth 5-10, regularization

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1. Bootstrap Confidence Intervals

Train each model **100 times** on bootstrap samples. Compute 95% CI. Parallel processing (**6**$\times$ speedup).

2. Permutation Importance Tests

Randomly permute features (**50** permutations). Measure drop in R^2. One-sample t-test ($\alpha = 0.05$).

3. Cross-Model Consistency

Spearman rank correlations between models. High $\rho > 0.7 \rightarrow$ robust rankings.

Goal

Ensure feature importance rankings are **statistically robust** and **consistent**

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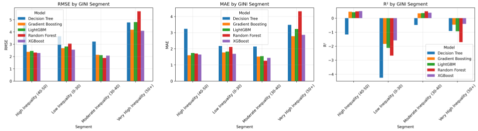
Testing context-dependence

By Income Level:

- Low income
- Lower-middle income
- Upper-middle income
- High income

By Geographic Region:

- East Asia & Pacific
- Europe & Central Asia
- Latin America & Caribbean
- Middle East & North Africa
- South Asia
- Sub-Saharan Africa



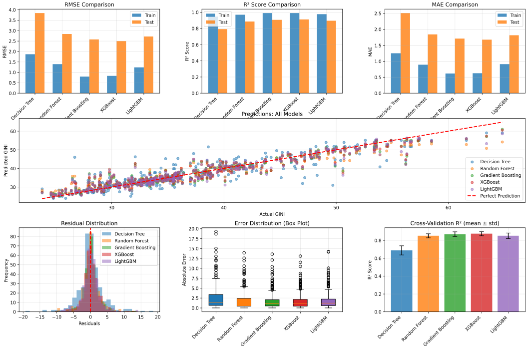
Key Question

Are inequality drivers **universal** or **context-specific**?

Model Performance: Ensemble Methods Excel

Results

Model	Test R^2	RMSE
Decision Tree	0.791	3.84
Random Forest	0.886	2.84
Gradient Boosting	**0.906**	2.58
XGBoost	**0.912**	**2.49**
LightGBM	0.895	2.72



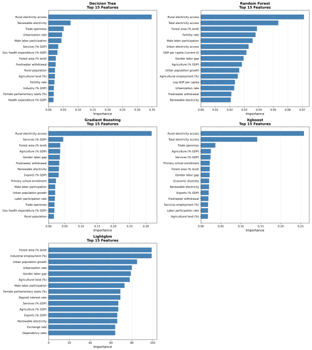
Key Findings

Ensembles **outperform** single tree. XGBoost & GB: $R^2 > 0.90$. Good generalization.

Top 10 Features: Infrastructure Dominates

Results

Rank	Feature	Imp.
1	Rural electricity	**0.258**
2	Total electricity	**0.141**
3	Trade openness	0.036
4	Agriculture % GDP	0.025
5	Services % GDP	0.024
6	Primary school enroll.	0.022
7	Forest area	0.022
8	Gender labor gap	0.022
9	Economic diversity	0.021
10	Renewable electricity	0.020

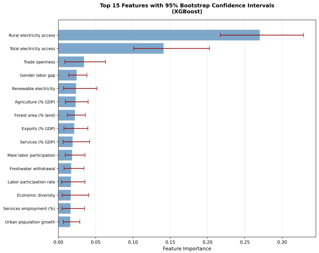


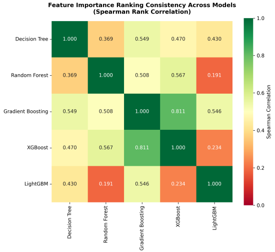
Key Implication

Inequality driven by **small subset** of critical factors

Statistical Significance Confirmed

Results





Bootstrap CI

All top 10: **95% CI** excludes zero

Cross-Model

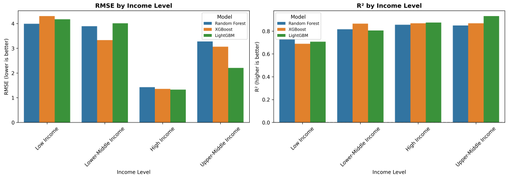
GB vs XGBoost: $\rho = 0.81$

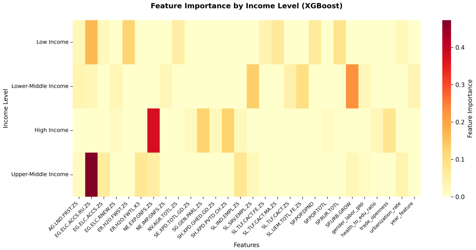
Conclusion

Top predictors **robust** across resampling, algorithms, and tests

Context Matters: Income-Level Heterogeneity

Results





Performance by Income Level

High: 0.86–0.87
Upper-mid: 0.85–0.93
Lower-mid: 0.81–0.87
Low: 0.69–0.73

Top Drivers Vary

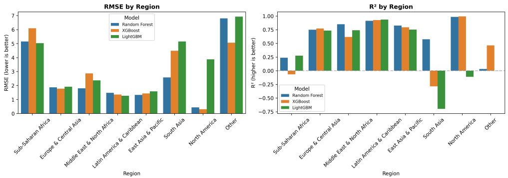
Low: Infrastructure
Mid: Urbanization
High: Trade & labor

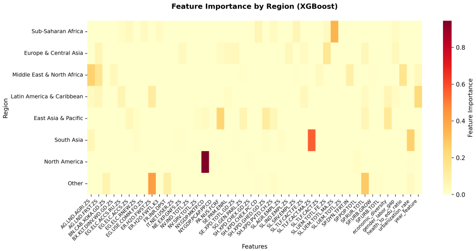
Policy Implication

Tailored interventions needed by development stage

Regional Heterogeneity: Geography Matters

Results





Performance by Region

Europe & C. Asia: Highest
Latin America: Moderate
Sub-Saharan Africa: Lower

Different Drivers

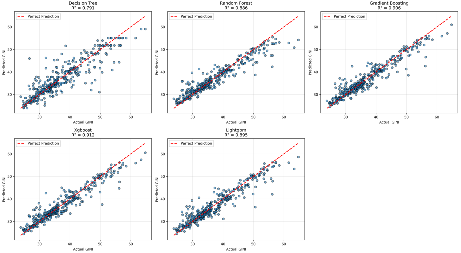
Region-specific factors matter

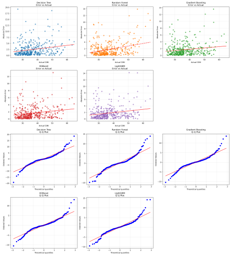
Key Insight

Inequality drivers vary by **income level** and **geography**

Model Diagnostics: Strong Performance

Results





Predicted vs Actual

Strong $R > 0.90$
GINI 20–45: excellent
GINI > 50: underprediction

Error Analysis

Errors near zero
Few outliers
Consistent across models

Conclusion

Models generalize well but struggle with extreme outliers

Computational Performance: 3–5× Speedup

Implementation & Optimization

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Parallel Processing Strategies

Component	Method	Before	After
Data collection	ThreadPoolExecutor (6 workers)	12 min	2 min
Bootstrap tests	joblib Parallel (n_jobs=-1)	60 sec	10 sec
Model training	scikit-learn parallelization	25 sec	5 sec
Total pipeline	Combined optimizations	**25–35 min**	**5–7 min**

Intelligent Caching

- SHA256 hash validation
- Auto cache invalidation
- **60–100**× speedup on reruns

Impact

- **6× faster** data collection
- **6× faster** bootstrap
- **5× faster** training

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Modular Architecture

9-stage pipeline: Data collection → Preprocessing → Training → Prediction → Evaluation → Comparison → Segmentation → Tests → Report

Execution Modes

- **Quick:** Full pipeline (default)
- **Fast:** 2015–2023 only
- **Optimized:** Hyperparameter tuning
- **Custom:** Fine-grained control

Quality Assurance

- Type annotations
- Comprehensive docstrings
- Data hash validation
- Deterministic results (seed=42)
- Dependency management

Reproducibility

- Git version control
- Automated caching
- Complete documentation
- Public code repository

Key Findings Summary

Conclusions & Policy Implications

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1. Predictive Power: Ensemble methods achieve $R^2 > 0.90$

- Inequality is highly predictable from socioeconomic factors
- XGBoost (0.912) & Gradient Boosting (0.906) perform best

2. Infrastructure Dominates: Rural electricity access (imp. 0.258)

- **1.8**$\times$ more important than next feature (total electricity: 0.141)
- Composite measure of development, institutions, connectivity

3. Robust Core Drivers: Trade, economic structure, gender gaps

- Consistent across all five models
- Statistically significant (bootstrap & permutation tests)

4. Context-Dependent: Different drivers at different income levels

- **Low income:** Basic infrastructure | **Middle:** Urbanization | **High:** Trade & labor markets

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Three priority areas for reducing inequality

1. Expand Infrastructure Access

- Particularly **rural electricity** connectivity
- Enables economic opportunities & education access
- Signals institutional capacity for service delivery

2. Promote Inclusive Labor Markets

- Reduce **gender labor gaps**
- Increase labor force participation (especially women)
- Address youth unemployment

3. Manage Structural Transformation

- Progressive taxation & social insurance
- Education access that scales with industrial demands
- Minimum wage floors to prevent wage compression

Limitations & Future Directions

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Limitations

- **Prediction not causation**
  - Need IV, DiD, RDD for causal effects
- Missing data imputation uncertainty
- Stationarity assumption
  - Relationships may change over time
- Omitted variables
  - Institutional quality, tax progressivity

Future Research

1. **Causal inference**
   - Double ML, causal forests
2. **Additional data**
   - Satellite imagery
   - Institutional quality measures
   - Alternative inequality metrics
3. **Methodological advances**
   - SHAP values for interactions
   - Panel methods with time-varying coefficients

Bottom Line

ML complements traditional econometrics: **prediction + pattern recognition** vs. **causal identification**

Contributions to Literature

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Methodological Contributions

- First systematic comparison of **5 tree-based ML algorithms**
- Robust statistical validation framework
- Advanced optimization (**3–5**$\times$ speedup)
- Fully reproducible pipeline

Innovation

Combines ML prediction power with rigorous statistical validation

Substantive Contributions

- **Infrastructure** as dominant predictor
- Context-dependence quantified
- High predictability ($R^2 > 0.90$)
- Tailored policy recommendations

Impact

Data-driven evidence for targeted inequality reduction

Data & Code: Publicly available for replication

THANK YOU

Any Questions?